INCH-POUND

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SUPERSEDING
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June 10, 1988

COMMERCIAL ITEM DESCRIPTION

THERMOMETER, SELF-INDICATING

LIQUID-IN-GLASS

The General Services Administration has authorized the use of this commercial item description in preference to Military Specification MIL-T-1180.

Abstract.

This description covers three types of liquid-in-glass, self-indicating thermometers graduated in Fahrenheit (°F) degrees, or Celsius (°C) degrees, for indicating tray and tank temperatures of photographic solutions.

Salient characteristics.

Design. The thermometer shall be liquid-in-glass, glass tube with the bulb at one end. The thermometer shall be mounted on a metal back.

Types. The thermometers shall be of the following types.

Type 1. Graduated in Fahrenheit degrees (°F).
Type 2. Graduated in Celsius degrees (°C).
Type 3. Graduated in both Fahrenheit and Celsius degrees (°F and °C.)

Style A - Thermosensitive fluid is mercury.
Style B - Thermosensitive fluid is not mercury.

Chemical Resistance. The thermometer scale marking, back plate, hook or fasteners shall show no evidence of chemical reaction when totally immersed in photographic chemicals.

Thermometer tube: The stem of the thermometer tube shall be marked in graduations from plus 40 °F to plus 140 °F, in 2 degree increments, for type 1 thermometer. Marking for the type 2 thermometer shall be from 4 °C to 60 °C in one degree increments. Marking for type 3 thermometer shall be the combined markings of type 1 and 2. Scale markings shall be legibly

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AMSC N/A FSC 6685

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etched on the thermometer stem. No scale markings shall be on the metal back plate. An arrow or heavy index mark shall be used to indicate the plus 68 °F mark, for type 1 and 20 °C for type 2. Graduation marks shall be clean cut, straight, and of uniform width. The width shall not exceed one-fifth the interval between graduations, and the separation between graduations shall be not less than 0.0625 inch. Graduation marks shall be clearly visible under normal darkroom conditions.

Fluid: The bulb and tube shall contain a sufficient quantity of thermosensitive fluid so that the height of the fluid in the tube shall indicate the temperature of the medium surrounding the thermometer within the range and accuracy specified herein.

Metal back: The thermometer shall be mounted on a corrosion-resisting steel back plate. The back plate shall extend at least the full length of the thermometer, and shall be curved to provide protection for the thermometer. The back shall have a hook of the same material as the back plate permanently attached at the top. The hook shall be designed so that when the thermometer is used in a developing tray, the hook will act as a standard for tilting the thermometer to a convenient angle for reading. When used in a developing tank, the hook shall enable the thermometer to hang freely from the edge of the tank.

Temperature range: The thermometer shall have a total temperature range of plus 40 °F to plus 140 °F, for type 1 and 4 °C to 60 °C for type 2. Type 1 thermometer shall indicate any temperature change within plus or minus 1.0 °F in a period of 30 seconds when totally immersed. Type 2 thermometer shall indicate any temperature change within plus or minus 0.5 °C in a period of 30 seconds when totally immersed. Type 3 shall be a combination of type 1 and type 2.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical. The offer/contractor will provide Material Safety Data Sheets of the fluid to the Government in accordance with FED-STD-313.

Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling and marking shall be specified in the contract or order.

Quality assurance. The thermometers shall be inspected for the following defects. Presence of one or more defects shall be cause for rejection of the item.

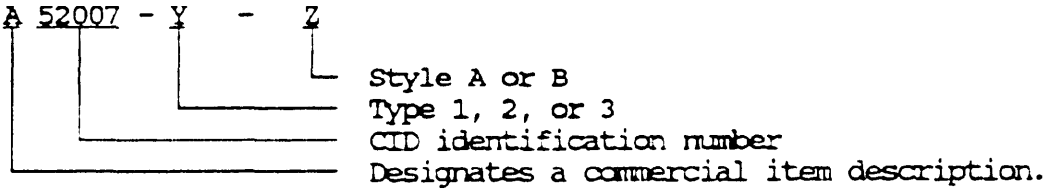
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1. Materials not as specified.
2. Types not as specified.
3. Dimensions not as specified.
4. Graduations not as specified.
5. Fluid not as specified.
6. Metal back not as specified.
7. Temperature range not as specified.
8. Chemical resistance not as specified.
9. Temperature accuracy testing at established intervals not as specified.

Certification of commercial item. The contractor shall certify that the thermometers offered conform to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product as the manufacturer's current product offered by the contractor in the commercial market place for at least one year preceding the solicitation and meets all requirements of this document as specified herein. The certification shall be furnished to the Government for approval prior to the first delivery and thereafter as specified in the contract/purchase order.

CID based part identification number. The following part identification umbering procedure is for government purposes and does not constitute a requirement for the contractor.

Part or identifying number (PIN). The PIN to be used for thermometers acquired to this commercial item description is created as follows:



Notes:

1. Ordering activity shall specify type and style of thermometer required.
2. When this commercial item description is used for procurement, the certification clause must appear in the solicitation.

Copies of FED-STD-313 and FED-STD-376 are available from the Naval Publications and Forms Center (attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

MILITARY INTERESTS:

Custodians
Army - ME
Air Force - 99

Review activities
Army - ER
DLA - GS

User activities
Air Force - 82

CIVIL AGENCY COORDINATING ACTIVITIES
GSA - FSS

PREPARING ACTIVITY:
Army - ME

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